

Research on Sorting System of Industrial Robot Based on Machine Vision

Yuntian Luo

Southwest Jiaotong University
Chengdu, China
2748156320@qq.com

Abstract—In the process of material sorting, the traditional manual sorting method is not only inefficient, but also has huge security risks. Affected by the types, shapes, quality and other factors of different materials, its sorting method also has some limitations. In order to solve the above problems, a material sorting system based on industrial robot is proposed in this paper. The hardware part of the sorting system mainly comprises an industrial camera, an industrial computer for processing image data, a control system, a robot control cabinet, an industrial robot, a frequency converter, a driving device and an executing mechanism. The software part of the sorting system includes a main module, a motion control module, a data communication module and an intelligent sorting module. This paper takes metal workpieces as the sorting object, and compares manual sorting with sorting system sorting. The results show that the designed sorting system can complete the sorting tasks of different sizes of metal workpieces in a short time, and the sorting efficiency and error rate are far lower than those of manual sorting. The industrial robot sorting system based on machine vision designed in this paper improves the industrial production efficiency and reduces the working cost.

Keywords—Sorting system, industrial robot, machine vision, industrial camera

I. INTRODUCTION

In order to solve the problem of low adaptability and flexibility of traditional sorting industrial robots, the concept of sorting industrial robots based on machine vision has been put forward. This is not only a research focus in the field of industrial automation, but also has great economic and practical significance. The sorting industrial robot obtains the image of the workpiece through the industrial camera, and uses a specific algorithm to extract the workpiece information in the image, such as shape, size, pose, etc. to achieve the purpose of simulating the human visual function, and then controls the mechanical arm to complete the sorting action, which significantly improves the ability of the industrial robot to recognize the surrounding environment information. Compared with the traditional industrial robots with manual teaching or off-line programming, the intelligent industrial robots with machine vision can adapt to more complex sorting working environment, and have higher flexibility and broader development and application prospects. The construction and improvement of industrial robot sorting system based on machine vision is based on program design, and through the application of machine vision technology, to achieve sorting and identification, and improve industrial production efficiency. Sorting is an important part of industrial production, but its process is relatively cumbersome, and in the logistics industry, sorting

is the largest part of the industrial chain that uses manual work. Therefore, in order to reduce the use of manpower and improve the industrial demand for automation, industrial robots are introduced to further improve the sorting level through automated sorting. In the process of studying the sorting process and behavior, it is necessary to improve the function and control of the sorting system through the joint application of automation technology and information technology. The application of machine vision in industrial robot sorting system is to improve the practical application effect of industrial robot sorting system from the perspective of vision technology and parcel sorting management.

This study takes the logistics parcel sorting robot as the main research object, and takes the continuous improvement of automation level as the premise, through the joint application of automation production mode and machine vision technology, to achieve the practical application effect of industrial robot sorting system. The first chapter is the introduction part, which introduces the background and significance of the research; the second chapter is to clarify the design requirements of the industrial robot sorting system; the third and fourth chapters elaborate the design points of the industrial robot sorting system based on machine vision from the hardware part and the software part respectively; In the fifth chapter, the running state of the industrial robot sorting system is tested and analyzed, and the experiment verifies that the sorting system can meet the design requirements. The sixth chapter is the conclusion.

II. LITERATURE REVIEW

D'Souza F, Costa J, Pires J N. studied the solution of integrating collaborative robots into industrial guided vehicles (AGVs), which has driven the development of industrial automation [1]. Shaikat A S, Akter S, and Salma U. have developed a computer vision based industrial robot arm that can classify and sort objects based on color and height [2]. Koskinopoulou M, Raptopoulos F, Papadopoulos G, et al. proposed a vision based recyclable waste classification system for industrial robots to separate garbage and improve resource recovery rates [3]. Chakole S, Ukani N. utilized low-cost vision systems and ABB industrial robots to achieve automatic picking and placing of items, reducing automation costs [4]. Zhang H, Liang H, Ni T, et al. studied a multi-objective sorting system based on deep learning, which improved the intelligence level of industrial robots [5]. Feng C, Nie G, Naveed Q N, et al. optimized the control system of sorting robots through deep learning and machine vision, improving sorting efficiency and accuracy [6]. Li M, Duan Y, He X, et al. proposed an image localization and recognition method and system for coal and gangue sorting robots, which

enhances the accuracy of sorting [7]. Babu K M C, Vardhini P A H. designed a low-cost item sorting system based on Arduino, providing a solution for resource limited environments [8]. Shang D, Wang Y, Yang Z, et al. studied the comprehensive calibration and image screening techniques of parallel robots for coal and gangue separation, which improved the sorting efficiency [9]. We can see that industrial robots have made significant progress in automation, intelligence, and cost-effectiveness. However, with the continuous growth of industrial automation demand, higher requirements have been put forward for the intelligence, flexibility, and cost-effectiveness of robot systems, which provides necessary background and impetus for the research in this article.

III. DESIGN METHODS

A. Design Method of Overall Hardware Architecture

The design of industrial robot sorting scheme needs to start from the overall framework structure, and analyze and study the hardware and software parts of the framework respectively. The hardware part includes industrial robot, flexible gripper, material input device, sensor and human-computer interaction interface. In terms of hardware design, the industrial robot is the core part of the material sorting system. The end effector generally uses flexible gripper, and the robot can successfully identify and grasp the target object within a certain error range. Its function is to tighten and hold the target object, and when the industrial robot moves to the designated position, it can be loosened and returned to the initial state. Generally speaking, the hardware part includes: industrial camera, industrial computer processing image data, control system, robot control cabinet, industrial robot, frequency converter, driving device and actuator. The overall hardware architecture of the system is shown in Fig. 1.

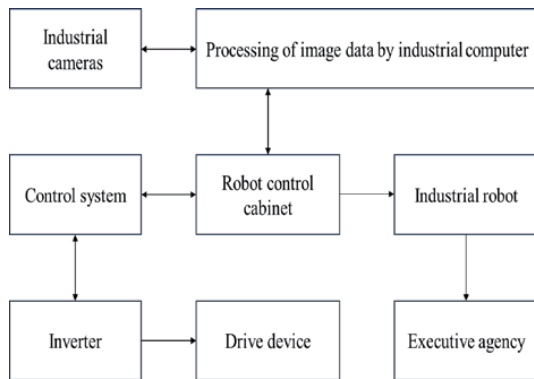


Fig. 1. Overall system hardware architecture

B. Design Method of Hardware Composition

Industrial robot. IRB2400 robot is selected as the most important hardware of the industrial robot sorting system in this paper. The robot is manufactured by ABB and its structure is shown in Fig. 2. It is available in a range of different configurations to meet the needs of a variety of applications, maximizing productivity in applications such as arc welding, machining, loading and unloading. The IRB2400 is a purpose-built, high-performance robot for machining applications with very demanding precision requirements. All models in this series can be installed

upside down. The IRB2400 has a compact design and is easy to install. Durable construction, reduced number of parts, high reliability, long maintenance intervals. In addition, the vibration and noise of its operation are small, the oil does not need to be replaced frequently, and the maintenance is simple. Moreover, IRB 2400 has a wide range of coverage, a long arm span and a relatively compact wrist structure, so it can still ensure accessibility in the application scenarios with more external obstacles or motion constraints. The main parameters of the IRB2400 robot are shown in TABLE I.



Fig. 2. IRB 2400 robot

TABLE I. MAIN PARAMETERS OF THE IRB2400 ROBOT

Parameter	Value
Load capacity	7kg
Reach distance	1.8m
Installation method	Floor and inverted installation
Additional load	Upper arm-wrist end 1 kg, rear end 10 kg, base 35 kg
Number of axes	The robot body 6, the outer shaft device 6,
Integrated signal source	23-pole, 50 VDC; 10-pole, 250VAC
Integrated air supply	8bar
Performance	Repeated positioning accuracy 0.06 mm (average value of ISO test)
Axis range of motion	Axis 1 rotated 360 °, Axis 2 arm 200 °, Axis 3 arm 125 °, Axis 4 wrist 370 °, Axis 5 bent 240 °, Axis 6 rotated 800 °
Speed	Axis 1 rotation 150 °/s, Axis 2 arm 150 °/s, Axis 3 arm 150 °/s, Axis 4 wrist 360 °/s, Axis 5 bent 360 °/s, Axis 6 rotation 450 °/s
Electrical connection	Supply voltage 200-600 V, 50/60HZ; Transformer rating 4kVA/7.8 kVA (with outer shaft)

Industrial Camera. The whole sorting system collects the image of the target object, and then distinguishes whether to sort and put into storage. In the whole industrial assembly line, the appearance shape of the target object in the sorting area is continuously photographed by a fixed camera position, and the acquisition and recognition of the target object are

completed through the processes of image processing, segmentation, recognition and the like, and whether the robot grasps the target object is judged. The Omron FZ-SC5M3 industrial camera is used in this paper. Omron OMRON captures images through the camera to achieve appearance, text, location, defects and other detection. A combination type image sensor having both an inspection capability and a rapid processing speed. Using AI, the previous visual detection relying on human eyes is automated, without defining and learning scratches, various scratches can be automatically detected, and deviation cognitive technology can be realized through AI. Scratches are found by a human-like "sensitivity"; it relies on the automation of the "sense detection" of the human sensibility. Visual inspection method for reproducing scratches by means of AI technology. FZ-SC5M3 industrial camera uses AI to automate the visual detection that used to rely on human eyes, with simple operation, up to 300000 pixels and short reading time. The Omron FZ-SC5M3 industrial camera is shown in Fig. 3.



Fig. 3. Omron FZ-SC5M3 industrial camera

Light source for the sorting area. The purpose of good target area lighting design is to make the image collected by the industrial camera can effectively distinguish the target object information from the background information. The application of the background light source reduces the difficulty of the segmentation preprocessing and the design of the target recognition algorithm of the image captured by the industrial camera. The accuracy, robustness, and various other properties of the industrial camera recognition system are also improved. If the light source is not designed correctly, the image processing algorithm and the performance of the image system will be difficult to achieve the desired results. The illumination components of backlight and the optical design of backlight are the preconditions for the design of machine vision recognition system. In the process of sorting system, the light source in the sorting area will directly affect the accuracy of the target object collected by the industrial camera. A good light source can improve the resolution of the whole image processing and reduce the difficulty of sorting. The sorting system uses led lights arranged in a circle, which can provide a large area of annular light source for the object to be measured. Its advantage is that it can be directly installed on the lens to achieve large area of lighting, and can better reflect the texture of the surface of the object to be measured.

C. Design Method of Software Framework

ABB robot system runs a modular Rapid language program built by Robot Studio platform, and uses the demonstrator of industrial robots to debug specific data such as position, input, output and communication. Industrial robots can operate autonomously according to pre-set procedures, thus completing a variety of complex production tasks. Through the debugging of specific data such as position, input, output and communication, engineers can accurately control and adjust the robot system to ensure its stability and reliability. The FZ image processing system running on the industrial computer provides important support for the automation of the industrial production line. After camera calibration, the system can collect images in real time, and calculate the plane center coordinates of the target object through the process image processing algorithm. These coordinate values can be fed back to the industrial robot control program through the communication cable, so as to realize the precise positioning and grasping of the target object by the robot. By combining with the PLC program to obtain the movement of the external axis, the industrial robot can flexibly adjust its own actions according to the real-time image information and the movement of the external axis, and realize the flexible grasping command module of the flexible end tool or axis softening. We can ignore the depth information of the object within a certain geometric size of the target object, and grasp and sort the target object. The software framework of the industrial robot sorting system is shown in Fig. 4.

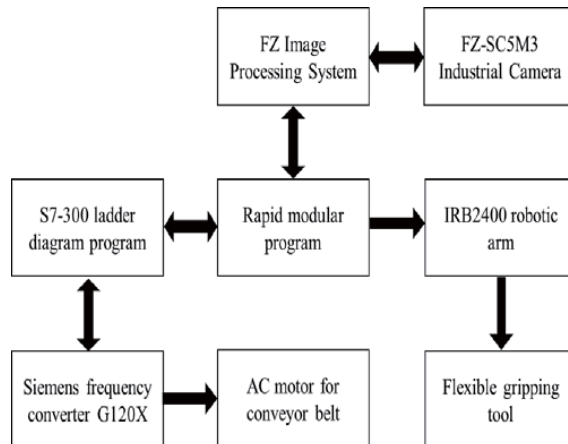


Fig. 4. Software framework of industrial robot sorting system

D. Design Method of Software Module

Main module. In order to ensure the normal operation of the sorting system, after the sorting robot and the system are powered on, the connection status of the system, the reset of the industrial robot and the rest of the gripper will be checked. After ensuring that there is no fault in the self-check, the robot will start to move from the starting position. After arriving at the sorting area, the machine vision module will be started to identify the target object. After successfully identifying the target object, the flexible gripper will pick up the target object, and then move the target object to the sorting area with the sorting robot. After arriving at the sorting area, the gripper will be released and return to the starting position. The cycle does not stop until the target has been completely sorted.

Motion Control Module. The design of this module mainly includes two kinds of trajectory control functions. It should be noted that, in order to facilitate us to realize the intelligent sorting function based on machine vision, the target point data not only includes the point data that the tool TCP needs to reach, but also needs to include additional information such as the safe picking and placing distance, the shape of the target, and the classification status of the target. Therefore, this module defines a special data type package. When the central point TCP of the robot end-effector moves in different working areas, some specific target points are needed as transition points to ensure that the collision can be avoided when the robot moves across areas, no matter what trajectory is taken [10].

Data communication module. The data communication module is mainly used to establish socket communication with the TCP/IP port of the Omron industrial computer FH-550L. From the perspective of network architecture, the upper layer of socket is the application process, and the lower layer is the TCP/IP network protocol, which can be said to be the interface between the application program for data interaction and the TCP/IP network protocol. The socket string sent from the industrial robot control cabinet IRC5 is mainly used to control the industrial camera for scene selection, tag selection, shooting parameter setting and shooting shutter control. The socket string returned from the industrial computer is mainly used to feed back the calculation results of the vision system, including the geometric shape classification of the target object, the geometric center coordinates of the target object, and the coordinate compensation value information.

Intelligent sorting module. The design requirements of the intelligent sorting module are mainly to collect the information of the target package, upload the collected information to the SQL database in the server for data comparison, and then store the data results returned by the database. The next placement area of the package is determined according to the classification data of the database [11]. Therefore, the design of this module includes the use of sockets for SQL database queries, as well as the storage and use of query results. The Omron industrial camera and the supporting industrial computer used in this system are equipped with Omron's visual processing software FZ-Panda, in which there are visual processing processes corresponding to bar code and two-dimensional code recognition. After parameter adjustment, the information carried by the bar code can be read through the Omron industrial camera [12]. According to different distribution destinations and package contents, the vision-based industrial robot sorting system is used to sort the packages into the subsequent intelligent warehousing.

IV. TEST OF INDUSTRIAL ROBOT SORTING SYSTEM

A. Experiment Result

In order to verify the function of the material sorting system of the industrial robot, 600 metal workpieces are taken as the sorting target, the metal workpieces are sorted on the conveyor belt according to the size, and compared with the manual sorting. The experiment results are shown in TABLE II.

TABLE II. COMPARISON OF INDUSTRIAL ROBOT SORTING AND MANUAL SORTING

Category	Sorting system		Manual sorting	
	Sorting quantity	Error rate	Sorting quantity	Error rate
Metal workpiece A (200)	198	1.0%	184	8.0%
Metal workpiece B (200)	195	2.5%	180	10.0%
Metal workpiece C (200)	191	4.5%	161	19.5%

The results show that the material sorting system based on industrial robot can greatly improve the sorting efficiency, and the sorting error rate is far lower than the manual sorting error rate, which verifies the effectiveness and reliability of the sorting system. In order to further improve the sorting efficiency and accuracy of the industrial robot material sorting system, the robot's motion speed and image recognition can also be debugged to reduce the sorting error rate and sorting time. In the traditional manufacturing industry, industrial robot sorting technology based on machine vision can be widely used in parts sorting, assembly, packaging and other links. By introducing intelligent sorting system, it can not only greatly improve production efficiency, but also reduce errors caused by manual operation and ensure product quality. At the same time, robot sorting technology can replace high-intensity and high-risk manual operation, reduce the labor intensity of workers and improve production safety.

B. Result Analysis

The robot sorting system has obvious advantages in improving production efficiency, as it can complete more work tasks in a shorter time, which means higher production capacity and faster market response speed for enterprises.

In traditional manufacturing, machine vision based industrial robot sorting technology can be widely applied in the picking, assembly, packaging and other processes of components. By introducing an intelligent sorting system, not only can production efficiency be greatly improved, but also errors caused by manual operations can be reduced, ensuring product quality. At the same time, robot sorting technology can replace high-intensity and high-risk manual operations, reduce the labor intensity of workers, and improve production safety. Secondly, in the field of agriculture, industrial robot sorting technology based on machine vision has broad application prospects in fruit and vegetable picking, agricultural product classification, and other links. In traditional agricultural production processes, picking and sorting often rely on a large amount of manpower and have low labor efficiency. By introducing intelligent sorting technology, precise identification and rapid classification of agricultural products can be achieved, production efficiency can be improved, the marketing cycle of agricultural products can be shortened, and labor costs can be reduced. Furthermore, in the field of logistics, industrial robot sorting technology based on machine vision is of great significance for improving the sorting efficiency of logistics centers and reducing cargo damage. The intelligent sorting

system can quickly and accurately identify packages of different shapes, sizes, and weights, achieve automated sorting, and greatly improve logistics efficiency. In addition, robot sorting technology can effectively reduce damage to goods during manual handling and lower logistics costs.

Although the industrial robot sorting system has shown high efficiency and accuracy in this experiment, there is still room for further optimization and improvement. By optimizing the motion path of robots, unnecessary actions and waiting time can be reduced, thereby further improving sorting speed. We can improve image recognition algorithms and increasing the accuracy of sensors, the occurrence of misidentification can be reduced and the error rate can be lowered.

V. CONCLUSIONS

With the progress and development of manufacturing industry, industrial robots play an increasingly important role in material sorting, which not only reduces the labor intensity, but also improves the production efficiency of enterprises, and is widely used in logistics, manufacturing and other fields. In this paper, an industrial robot sorting system based on machine vision is proposed, which uses machine vision technology to recognize and classify the objects on the industrial production line, and realizes automatic sorting through industrial robots. Through the analysis of the hardware design, software design and experimental results, the feasibility and effectiveness of the system are verified. The system has the advantages of high automation, high sorting speed and high accuracy, and has broad application prospects.

REFERENCES

- [1] D'Souza F, Costa J, Pires J N. Development of a solution for adding a collaborative robot to an industrial AGV[J]. *Industrial Robot: the international journal of robotics research and application*, 2020, 47(5): 723-735.
- [2] Shaikat A S, Akter S, Salma U. Computer vision based industrial robotic arm for sorting objects by color and height[J]. *Journal of Engineering Advancements*, 2020, 1(04): 116-122.
- [3] Koskinopoulou M, Raptopoulos F, Papadopoulos G, et al. Robotic waste sorting technology: Toward a vision-based categorization system for the industrial robotic separation of recyclable waste[J]. *IEEE Robotics & Automation Magazine*, 2021, 28(2): 50-60.
- [4] Chakole S, Ukani N. Low-cost vision system for pick and place application using camera and ABB industrial robot[C]//2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT). IEEE, 2020: 1-6.
- [5] Zhang H, Liang H, Ni T, et al. Research on multi-object sorting system based on deep learning[J]. *Sensors*, 2021, 21(18): 6238.
- [6] Feng C, Nie G, Naveed Q N, et al. Optimization of sorting robot control system based on deep learning and machine vision[J]. *Mathematical Problems in Engineering*, 2022(1): 1-7.
- [7] Li M, Duan Y, He X, et al. Image positioning and identification method and system for coal and gangue sorting robot[J]. *International Journal of Coal Preparation and Utilization*, 2022, 42(6): 1759-1777.
- [8] Babu K M C, Vardhini P A H. Design and development of cost effective arduino based object sorting system[C]//2020 International conference on smart electronics and communication (ICOSEC). IEEE, 2020: 913-918.
- [9] Shang D, Wang Y, Yang Z, et al. Study on comprehensive calibration and image sieving for coal-gangue separation parallel robot[J]. *Applied Sciences*, 2020, 10(20): 7059.
- [10] Thuyet D Q, Kobayashi Y, Matsuo M. A robot system equipped with deep convolutional neural network for autonomous grading and sorting of root-trimmed garlics[J]. *Computers and Electronics in Agriculture*, 2020, 178: 105727.
- [11] Zhang F, Lin Y, Zhu Y, et al. A Real-Time Sorting Robot System for Panax Notoginseng Taproots Equipped with an Improved Deeplabv3+ Model[J]. *Agriculture*, 2022, 12(8): 1271.
- [12] Bader F, Rahimifard S. A methodology for the selection of industrial robots in food handling[J]. *Innovative Food Science & Emerging Technologies*, 2020, 64: 102379.
- [1] D'Souza F, Costa J, Pires J N. Development of a solution for adding a collaborative robot to an industrial AGV[J]. *Industrial Robot: the*